

LIBRARY MANAGEMENT SYSTEM – ISSUES ANALYSIS DOCUMENT

OVERVIEW

A comprehensive review of the Library Management System identified multiple critical, major, and minor issues affecting functionality, maintainability, performance, and code quality. The system contains problems related to JavaScript fundamentals, object-oriented programming, DOM manipulation, event handling, testing, modern ES6+ practices, and error handling.

CRITICAL ISSUES

1. Undeclared global variable `books` causing global scope pollution.
2. Undeclared constant `MAX_BOOKS_PER_MEMBER`.
3. Extensive use of `var` instead of `let` and `const`.
4. Missing `super()` call in `DigitalBook` constructor causing inheritance failure.
5. Assignment operator (`=`) used instead of comparison operator in `Member.canBorrow()`.
6. Assignment operator (`=`) used in `searchBooksByCategory()`.
7. Assignment operator (`=`) used in `findMemberById()`.
8. Assignment operator (`=`) used in `handleFilterChange()`.
9. Infinite loop in `processReturnQueue()` due to missing index increment.
10. Recursive function `searchBooksByCategory()` lacks a base case causing stack overflow.
11. Missing null and undefined checks in recursive functions.
12. Missing validation in `borrowBook()` may cause runtime errors.
13. Missing availability validation in `Book.checkOut()`.
14. `importLibraryData()` can crash on invalid JSON.
15. `loadFromLocalStorage()` stores strings directly instead of parsed objects.
16. `saveToLocalStorage()` stores arrays without `JSON.stringify()`.
17. UI initialized before DOM is loaded.
18. Missing null checks on DOM elements before manipulation.
19. `findBookByISBN()` and `findMemberById()` may return invalid values without handling.
20. Tests fail because required test data is never initialized.

MAJOR ISSUES

Variables and Operators

- Multiple uses of `var` throughout the application.
- Use of `==` instead of `===`.
- Missing type validation.
- Missing NaN protection in calculations.

Control Flow

- Inefficient nested loops in `findOverdueBooks()`.

- Missing edge-case handling.
- Poor loop selection in several functions.
- Lack of proper return handling.

Functions and Functional Programming

- `getBooksByAuthor()` should use `filter()`.
- `calculateTotalLateFees()` should use `reduce()`.
- `findMemberById()` should use `find()`.
- `combineBookCollections()` should use spread syntax.
- `addMultipleBooks()` should use rest parameters.
- `updateMemberInfo()` should use object destructuring.
- Missing higher-order function implementations.
- Missing pure function implementations.

Object-Oriented Programming

- Book class missing `availableCopies` property.
- Book class missing `totalCopies` property.
- Book class missing `isAvailable()` method.
- Book class missing `getInfo()` method.
- Member class missing `joinDate` property.
- Member class missing membership duration calculation.
- Member class missing destructuring-based information method.
- `PremiumMember` class does not override `canBorrow()`.
- `PremiumMember` missing premium-specific benefits.
- `LibraryStats` missing required statistics methods.

Modern JavaScript

- Missing template literals.
- Missing destructuring.
- Missing spread operator usage.
- Missing rest parameter usage.
- Missing ES6 module exports.
- Missing ES6 module imports.

DOM and Event Handling

- Incorrect selector syntax for filter dropdown.
- Missing search input listener.
- Wrong event type used for filtering.
- Missing event delegation.
- Missing `preventDefault()` in form submissions.
- Catalogue not cleared before rendering.
- Missing data attributes for dynamic elements.

- Case-sensitive search implementation.
- Poor DOM update performance.

Testing

- Incomplete Book tests.
- Missing DigitalBook tests.
- Missing PremiumMember tests.
- Missing recursion tests.
- Missing error-handling tests.
- Missing DOM tests.
- Missing JSON tests.
- Missing localStorage tests.
- Missing edge-case coverage.
- Coverage significantly below required threshold.

MINOR ISSUES

1. String concatenation used instead of template literals.
2. Inconsistent coding style.
3. Lack of comments and documentation.
4. Missing user feedback for failed operations.
5. Missing form reset after submission.
6. Missing placeholders and labels in forms.
7. Incorrect email input type handling.
8. Poor naming consistency.
9. Repeated DOM updates reducing performance.
10. Missing currency formatting for fines.
11. Missing string normalization methods.
12. Missing use of DocumentFragment for rendering.

MISSING FEATURES

1. Book availability tracking.
2. Membership duration calculations.
3. Premium borrowing benefits.
4. Enhanced library statistics.
5. JSON export and import validation.
6. LocalStorage persistence system.
7. Search functionality improvements.
8. Category filtering improvements.
9. Error handling infrastructure.
10. ES6 module architecture.
11. Comprehensive Jest test suite.
12. Coverage reporting and validation.

PROPOSED FIX STRATEGY

Phase 1:

Fix variable declarations, operators, comparisons, and scope issues.

Phase 2:

Repair loops, conditionals, recursion, and control flow problems.

Phase 3:

Complete Book, DigitalBook, Member, and PremiumMember implementations.

Phase 4:

Refactor functions using filter, map, reduce, find, some, and every.

Phase 5:

Modernize code using destructuring, template literals, spread, and rest operators.

Phase 6:

Implement robust error handling, validation, and edge-case protection.

Phase 7:

Repair DOM interactions, event listeners, JSON handling, and localStorage functionality.

Phase 8:

Develop comprehensive Jest test coverage and documentation.

Total Issues Identified: 50+

Priority Order:

Critical → Major → Minor

This approach minimizes application-breaking defects first and ensures all rubric requirements can be satisfied systematically.